

QUANTUM INFORMATION THEORY

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1. COPENHAGEN INTERPRETATION OF QUANTUM MECHANICS

The theory of Quantum Mechanics emerged from a series of experiments that made many physicists, around the beginning of the 20th century, understand that energy in nature does not present itself in a *continuum spectrum* but rather in discretized levels, these being called by Planck: *quanta*. Energy can only take integer multiple of a specific fundamental value: Planck's constant $h \approx 6.6 \times 10^{-34}$. Over the 20's, two main formulations rose, both with the aim of describing a quantum system via a *characteristic value problem*, that is, an *eigenvalue problem*: Schrödinger's one, involving some differential operator $\mathcal{O}$,

$$\mathcal{O}\psi = \lambda\psi$$

and Heisenberg's one, involving simply a matrix M ,

$$Mv = \lambda v$$

where wavefunctions are treated as simple vectors. We speak of *wavefunction* because, given coordinates q the associated *canonical momentum* is the differential operator $q = k \partial/\partial q$ where $k = i\hbar$ is chosen so that, as $\hbar \equiv h/(2\pi) \downarrow 0$, Schrödinger's equation

$$\mathcal{H}\psi = i\hbar \frac{\partial\psi}{\partial t}$$

resembles the Hamilton-Jacobi equation in the classical limit and, formulated as

$$\mathcal{H}\psi = E\psi$$

so that energy levels are just the eigenvalues of the Hamiltonian operator, yields

$$i\hbar \frac{\partial\psi}{\partial t} = E\psi$$

which has as general solution the wave

$$\psi(x, t) = \psi(x, 0) \exp\left(-i \frac{E}{\hbar} t\right).$$

However, the scope of the theory was to model particles, in fact the very motivation behind quantum theory is to explain cumbersome experiments (e.g. Young's double-slit experiment) involving electrons. It was Born who first proposed the interpretation of wavefunctions as giving the probability of the electron being in some position via its magnitude squared:

$$d\wp = |\psi(x)|^2 dx, \quad 1 = \int_{\text{all } x} |\psi(u)|^2 du$$

rather than full deterministic information about the particle's exact position. This means that wavefunctions need to be square-integrable, i.e. live in a (infinite-dimensional, Hilbert) vector space, and Heisenberg's picture $Mv = \lambda v$ makes sense. To unify these two 'pictures', Dirac proposed a formalism where a vector space spanned by some abstract '*ket*' unitary statevectors $|n\rangle \cong \hat{e}_n$ are considered, and their '*bra*' duality

pairings $\langle m|$ so that their dot product is the scalar ‘bra-ket’ $\langle m|n\rangle = \delta_{m,n}$, this corresponding to the square-integrability condition

$$\int \psi_m^* \psi_n = \delta_{m,n}$$

Correspondingly, operators are of the form $|\dots\rangle\dots\langle\dots|$.

2. QUANTUM COMPUTING

Classical bits are $b = 0, b = 1$, while the quantum bit, or **qubit**, is

$$|q\rangle = a|0\rangle + b|1\rangle \quad \text{s.t. } |a|^2 + |b|^2 = 1$$

A quantum system with n qubits contains the information of 2^n complex numbers, so modeling such a system with a classical computer will require storage of a vast amount of floating point data. Consider system number one with basis $|0\rangle, |1\rangle$ with operator $\mathcal{O}_1$ acting on it alone, and system number two with basis $|a\rangle, |b\rangle$ with operator $\mathcal{O}_2$ acting on it alone. We can create a four state system using **abstract tensors** whose basis is

$$|0, a\rangle \equiv |0\rangle \otimes |a\rangle, \quad |1\rangle \otimes |a\rangle, \quad |0\rangle \otimes |b\rangle, \quad |1\rangle \otimes |b\rangle$$

and extend the action of $\mathcal{O}_1, \mathcal{O}_2$ to actions on these states via

$$\mathcal{O}_1 \rightarrow \mathcal{O}_1 \otimes 1, \quad \mathcal{O}_2 \rightarrow 1 \otimes \mathcal{O}_2$$

i.e.

$$\mathcal{O}_1 \otimes 1(|0\rangle \otimes |a\rangle) \equiv (\mathcal{O}_1|0\rangle) \otimes (1|a\rangle) = (\mathcal{O}_1|0\rangle) \otimes (|a\rangle)$$

$$1 \otimes \mathcal{O}_2(|0\rangle \otimes |a\rangle) \equiv (1|0\rangle) \otimes (\mathcal{O}_2|a\rangle) = (|0\rangle) \otimes (\mathcal{O}_2|a\rangle)$$

For example, the *single-particle* Hamiltonian $\mathcal{H}|i\rangle = E_i|i\rangle$ with $i = 1, 2$, can be linearly extended to act on the 2-qubit system:

$$\begin{aligned} \mathcal{H}|i, j\rangle &= \mathcal{H}|i\rangle \otimes \mathcal{H}|j\rangle \\ &= \mathcal{H}|i\rangle \otimes |j\rangle + |i\rangle \otimes \mathcal{H}|j\rangle \\ &= E_i|i\rangle \otimes |j\rangle + |i\rangle \otimes E_j|j\rangle \\ &= (E_i + E_j)|i\rangle \otimes |j\rangle \end{aligned}$$

by the transformation

$$\mathcal{H} \longrightarrow \mathcal{H} \otimes 1 + 1 \otimes \mathcal{H}.$$

We can now create what are called **gates in quantum computing**, which are basically unitary transformations made up of qubits, namely, a transformation $\mathcal{U}$ such that $\mathcal{U}\mathcal{U}^\dagger = \text{Id} = \mathcal{U}^\dagger\mathcal{U}$ which correspond to *isometries*

$$\langle \mathcal{U}\psi | \mathcal{U}\phi \rangle = \langle \psi | \mathcal{U}^\dagger \mathcal{U} | \phi \rangle = \langle \psi | \phi \rangle$$

that thus preserve both distances and lengths.

A *Hadamard gate* is an example of a unitary transformation of a statevector

$$|0\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |1\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

which takes a single qubit and *spits out* a linear combination of qubits. So, its matrix representation (acting on the matrix representation of the qubits) is:

$$\text{Rep}(\text{HD}) = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}$$

3. ENTANGLEMENT

We say that quantum states are **entangled** if they are not factorizable. For example, the state $|0\rangle \otimes |1\rangle$ is not entangled, same for

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{5}}(2|0\rangle - |1\rangle)$$

The **Bell states**

$$|B_1\rangle = \frac{|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle}{\sqrt{2}}$$

$$|B_2\rangle = \frac{|0\rangle \otimes |0\rangle - |1\rangle \otimes |1\rangle}{\sqrt{2}}$$

$$|B_3\rangle = \frac{|0\rangle \otimes |1\rangle + |1\rangle \otimes |0\rangle}{\sqrt{2}}$$

$$|B_4\rangle = \frac{|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle}{\sqrt{2}}$$

cannot be factored and are thus entangled. The tensor product of the factor of first particle with the factor for the second:

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle$$

is not instead entangled.

3.1. **Density.** We can define the **density operator**

$$\rho = |\psi\rangle\langle\psi|$$

carrying the same amount of information that the wavefunction does. Expressed in a specific basis, the density matrix is actually a collection of operators:

$$\rho = \sum_{n,m} \rho_{n,m} |n\rangle\langle m|$$

and there will a unitary transformation that, when performed, will allow us to express the density matrix in a basis in which it can be diagonalized:

$$\rho = \sum_i \lambda_i |i\rangle\langle i|$$

We define the **trace** of an operator as follows:

$$\text{Tr}|\psi\rangle\langle\chi| = \langle\chi|\psi\rangle$$

We can carry out measurements with the density matrix: for example, the probability that the system will collapse into the state j , which is also one of the states in the basis in which ρ is diagonalized, i.e., the probability that, when measuring eigenvalues, the outcome is λ_j , is:

$$\wp(j) = \text{Tr}(\rho|j\rangle\langle j|) = \langle j|\rho|j\rangle$$

is the eigenvalue itself, hence **the eigenvalues of the density matrix are probabilities!** Hence

$$\text{Tr}\rho = \text{Tr} \sum_i \lambda_i |i\rangle\langle i| = \sum_i \lambda_i \langle i|i\rangle = \sum_i \lambda_i = 1.$$

The **reduced density matrix** for, say, a two-component system is what we would obtain by *integrating out* all of the quantum degrees of freedom of one of the two systems and leaving behind just something that will tell you about measurements outcomes on one of the systems only.

Suppose starting out with $\rho = |\psi\rangle\langle\psi|$ and *sum/average out* all the information about the states of the second system of the wavefunction:

$$\rho_1 = \text{Tr}_2 |\psi\rangle\langle\psi|$$

containing information about possible outcomes of measurements made on particle number 1.

Suppose having an universe composed of an enormous number of particles N , then we can trace out the states of all quantum particles except the first:

$$\rho_1 = \text{Tr}_{2,\dots,N} |\psi\rangle\langle\psi|$$

leaving us with a single quantum system entangled with an enormous number of other and this allows to reduce us to a single-system quantum mechanics on a multi-component system wavefunction.

3.2. Conditional probabilities. Start with $|\psi\rangle$ and consider a measurement with $\mathcal{J}$, with eigenspace $\{|j_n\rangle\}$. This is a complete basis of $\mathcal{H}$ so that

$$|\psi\rangle = \sum_n \psi_n |j_n\rangle \quad \text{alternatively} \quad \rho = \sum_n \wp_n |j_n\rangle\langle j_n|$$

Consider the operator $\mathcal{A}$, with eigenspace $\{|a_m\rangle\}$. We can consider the conditional probability that a state $|j_n\rangle$ will be found to have a $\mathcal{A}$ -eigenvalue a_m :

$$\wp(a_m|j_n) = |\langle a_m|j_n\rangle|^2$$

and suppose the density matrix is diagonal in this basis:

$$\rho = \sum_n \wp_n |j_n\rangle\langle j_n|$$

where eigenvalues are the probabilities that, measuring $\mathcal{J}$ on the superposition for which ρ is the density, one gets j_n , hence the total probability

$$\begin{aligned} \wp(a_m) &= \sum_n \wp_n \wp(a_m|j_n) = \sum_n \wp_n \langle j_n|a_m\rangle\langle a_m|j_n\rangle \\ &= \text{Tr} \left(|a_m\rangle\langle a_m| \sum_n \wp_n |j_n\rangle\langle j_n| \right) = \text{Tr}(|a_m\rangle\langle a_m|\rho) \end{aligned}$$

and we built a full measurement theory in which all calculations make no reference to wavefunctions whatsoever! The density matrix ρ gives another way of describing a quantum system that can't readily be described by ψ : there is another way of doing quantum mechanics that does not require wavefunctions! Suppose the density matrix is idempotent:

$$\rho^2 = \rho$$

then the system described by this density matrix is in a **pure state**, that is, it is described nicely by a single wavefunction and there exists $|\psi\rangle$ s.t. ρ factors, exactly as we have defined it; in fact, in such case:

$$\rho^2 = |\psi\rangle\langle\psi|\psi\rangle\langle\psi| = |\psi\rangle \cdot 1 \cdot \langle\psi| = |\psi\rangle\langle\psi| = \rho$$

If $\rho^2 \neq \rho$, the system is in a **mixed state**. It is possible to describe a mixed state with a statevector embedded in a larger space by entangling it with an environment, so that its reduced density matrix is not idempotent, but this is not really necessary, as we can hypothesize quantum systems described by such a density matrix and have a full measurement theory for them without having to mess with wavefunctions at all, and this may of course be useful for systems so tightly entangled with another system (environment/lots of other particles or an instrument), that we cannot dis-entangle them.

3.3. **Entropy of entanglement.** There are two equivalent measures of entanglement:

- The **Gini Impurity Index**

$$\text{GI}(p) = \sum_i \lambda_i (1 - \lambda_i)$$

- The **Entropy**

$$S_E(p) = - \sum_i \lambda_i \log \lambda_i$$

which is here to be interpreted, as for the information-theoretic Shannon entropy

$$H = - \sum_i p_i \log p_i$$

of a statistical ensemble, as a **measure of surprise**: if an outcome that has almost zero probability occurs, then it has a very entropy, while if an event has a very high probability of occurring, then we are not at all surprised and the presence of the logarithm makes the entropy vanish. Eventually, we will show that this corresponds exactly to Gibbs' thermodynamical entropy, for now is enough to think of the early universe as a pure un-entangled quantum state with no entropy. Then, as time goes by and things get *more and more complicated and disordered*, there is a good reason to argue that the degree of entanglement increases without bound.

E.g., given $\rho_2 = \text{Tr}_1 |B_1\rangle\langle B_1| = (|0\rangle\langle 0| + |1\rangle\langle 1|) / 2 = 1/2 \text{Id}_2$,

$$S_E(p) = - \left(\frac{1}{2} \log \frac{1}{2} + \frac{1}{2} \log \frac{1}{2} \right) = \log 2 = 1$$

hence each qubit of the bell state is maximally entangled.

If instead, we start with the pure product

$$|\psi\rangle = (\cos \theta |0\rangle + \sin \theta |1\rangle) \otimes |0\rangle$$

which is not entangled, then

$$\begin{aligned} \rho_1 &= \text{Tr}_2 |\psi\rangle\langle\psi| = \text{Tr}_2 (\cos^2 \theta |0,0\rangle\langle 0,0| + \sin \theta \cos \theta |0,0\rangle\langle 1,0| \\ &\quad + \cos \theta \sin \theta |1,0\rangle\langle 0,0| + \sin^2 \theta |1,0\rangle\langle 1,0|) \\ &= (\cos^2 \theta |0\rangle\langle 0| + \sin \theta \cos \theta |0\rangle\langle 1| + \cos \theta \sin \theta |1\rangle\langle 0| + \sin^2 \theta |1\rangle\langle 1|) \end{aligned}$$

with

$$\text{Rep}(\rho_1) = \begin{pmatrix} \cos^2 \theta & \sin \theta \cos \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{pmatrix}$$

and eigenvalues $\lambda = 0, 1$. This is a general fact: **for an un-entangled wavefunction, the eigenvalues of either of the two reduced densities are 0, 1 and thus the entropy is 0:**

$$S_E = \lim_{x \uparrow 0} -x \log x = 0, \quad \text{GI} = 0(1+0) + 1(1-1) = 0$$

There is no such thing as a *partial entanglement*, but there is a continuous spectrum for the degree of entanglement of a system, e.g.

$$|\psi\rangle = \cos \theta |0\rangle \otimes |1\rangle + \sin \theta |1\rangle \otimes |0\rangle$$

has

$$\rho_1 = \text{Tr}_2 |\psi\rangle\langle\psi| = \cos^2 \theta |0\rangle\langle 0| + \sin^2 \theta |1\rangle\langle 1| = \begin{pmatrix} \cos^2 \theta & 0 \\ 0 & \sin^2 \theta \end{pmatrix}$$

and we can find θ values for which the entropy

$$S_E = - \cos^2 \theta \log \cos^2 \theta - \sin^2 \theta \log \sin^2 \theta$$

vanishes or show that the maximally entangled version is one of the Bell states, by extremizing the entropy functional.

4. QUANTUM INFORMATION THEORY

Classical Shannon's entropy $H(p) = -\mathbb{E}_p[\log p]$ can easily be extended to the case of a (diagonalized) density matrix $\rho = \sum_i p_i |i\rangle\langle i|$ describing our quantum system, rather than some probability density p , by defining $\log \rho \equiv \sum_i \log(p_i) |i\rangle\langle i|$ and taking the analogous of an expectation by tracing the latter. This is **Von Neumann's entropy**

$$H(\rho) = -\text{Tr } \rho \log \rho$$

which is nonnegative, achieves its maximum value for the maximally mixed state, and is zero if and only if ρ is pure: the minimum value occurs when the eigenvalues of a density operator are distributed with all the probability mass on one eigenvector and zero on the others, so that the density operator is rank one and corresponds to a pure state. Concavity also follows from concavity of Shannon's entropy: entropy can never decrease under a mixing/entangling operation.

However, the most important property is that of isometric invariance (under unitary transformations):

$$H(\mathcal{U}\rho\mathcal{U}^\dagger) = H(\rho)$$

as a unitary evolution takes pure states into pure states, and it is essentially a change of basis which does not change the entanglement configuration of the quantum system, and, mathematically, does not change the eigenvalues.

We can also define the quantum joint entropy for two subsystems described by states ρ_A and ρ_B of a composite system described by a joint state ρ_{AB} as follows:

$$H(A, B) = H(\rho_{AB})$$

and the quantum conditional entropy

$$\begin{aligned} H(A|B) &\equiv H(A, B) - H(B) \\ &= H(\rho_{AB}) - H(\rho_B). \end{aligned}$$

However, unlike the classical conditional entropy, the quantum conditional entropy can become negative, indicating that it is possible for quantum systems to be more certain in the joint state of two component systems than in the states of its individual components.

The first bell state itself, taken as a whole, is pure, in fact the joint $\rho = |B_1\rangle\langle B_1|$ has zero entropy and thus, letting A and B be respectively the (maximally) entangled reduced densities ρ_1 and ρ_2 :

$$H(A|B) = H(\rho) - H(\rho_2) = -1$$

in fact, each qubit is entangled, but all the uncertainty about its collapse vanishes by observing the second qubit. Einstein called them "spooky" because two of such systems determine one another's measurement instantaneously, and no information is passed! In fact, If we measure the first qubit and obtain $|0\rangle$, the system collapses to $|0, 0\rangle$, meaning the second qubit is also certainly in $|0\rangle$.

4.1. Quantum relative entropy. The quantum relative entropy between the two states, ρ and σ , of a quantum system is defined as

$$D(\rho\|\sigma) \equiv \text{Tr } \rho(\log \rho - \log \sigma)$$

It is monotone and obeys *Klein's inequality*

$$D(\rho\|\sigma) \geq 0$$

which is the analogous to Gibbs' inequality for the Kullback-Leibler divergence. The proof, supposing $\rho = \sum_i \rho_i |i\rangle\langle i|$ and $\sigma = \sum_j \sigma_j |j\rangle\langle j|$, takes one line:

$$D(\rho\|\sigma) = \sum_i \rho_i \left(\log \rho_i - \sum_j \log \sigma_j \langle i|j\rangle^2 \right) = \sum_i \rho_i \log \frac{\rho_i}{\sigma_j} \geq 0$$

where we conclude thanks to Gibbs, and get equality if and only if $\rho = \sigma$.

The quantum mutual information between two subsystems is

$$\begin{aligned} I(A, B) &= D(\rho_{AB} \| \rho_A \otimes \rho_B) \\ &= H(\rho_A) + H(\rho_B) - H(\rho_{AB}). \end{aligned}$$

where we the tensor product $\rho_A \otimes \rho_B$ represents the quantum universe as if A and B were independent.

4.2. Quantum Jensen-Shannon divergence. The Jensen-Shannon divergence

$$\text{JD}(p, q) = H\left(\frac{p+q}{2}\right) - \frac{1}{2}(H(p) + H(q))$$

is a symmetrized and smoothed version of the KL divergence. As opposed to the latter, it determines in a very direct way a metric; indeed, it is the square of the so-called *transmission metric* d_T . A natural way to extend Jensen-Shannon divergence is to consider a mixture of k probability distributions $p_1, \dots, p_k$, with weights $\pi_1, \dots, \pi_k$, respectively, and consider

$$\text{JD}(p_1, \dots, p_k) \equiv H\left(\sum_{i=1}^k \pi_i p_i\right) - \sum_{i=1}^k \pi_i H(p_i)$$

which is immediately quantizable

$$\text{QJD}(\rho_1, \dots, \rho_k) \equiv H\left(\sum_{i=1}^k \pi_i \rho_i\right) - \sum_{i=1}^k \pi_i H(\rho_i).$$

5. QUANTUM COMMUNICATION

A quantum channel is a means for transmitting quantum information. Physically, a quantum channel may be viewed as a medium of transmission, such as an optical fiber, together with an ensemble of quantum systems, such as photons, prepared by the sender in quantum states ρ_i , $i = 1, 2, \dots, n$, carrying n symbols with corresponding probabilities p_i . Due to the inevitable, generally irreversible, interaction of a transmitted quantum system with the environment of a realistic quantum channel, however, quantum channels are fundamentally different in character from classical channels, just as qubits differ from bits.

Consider a quantum channel through which a pure quantum state $|\psi\rangle$ can be sent, then its *fidelity* is the expectation $\langle \psi | \rho | \psi \rangle$ where now ρ is the state of the system after transmission, as a measure of channel faithfulness; a quantum channel is *faithful* if this expectation value goes to unity in the appropriate information-processing limit.

5.1. Holevo's Theorem. Let A be the (classical) input, ρ the receiver's measurement result of the quantum system Q and B the (classical) output, then

$$I(A, B) \leq \text{QJD}(\rho)$$

and the following calculation is a proof:

$$\begin{aligned}
I(A, B) &\leq I(A, Q) = H(A) + H(\rho) - \text{Tr}(\rho_{AB} \log \rho_{AB}), \quad \rho_{AB} = \sum_{i=1}^n p_i |i\rangle\langle i| \otimes \rho_i \\
&= H(A) + H(\rho) + \text{Tr} \left(\sum_{i=1}^n p_i \log p_i |i\rangle\langle i| \otimes \rho_i \right) + \text{Tr} \left(\sum_{i=1}^n p_i |i\rangle\langle i| \otimes \rho_i \log \rho_i \right) \\
&= H(A) + H(\rho) + \text{Tr} \left(\sum_{i=1}^n p_i \log p_i |i\rangle\langle i| \right) + \text{Tr} \left(\sum_{i=1}^n p_i \rho_i \log \rho_i \right) \\
&= H(A) + H(\rho) - H(A) + \sum_{i=1}^n p_i \text{Tr}(\rho_i \log \rho_i) = \text{QJD}(\rho).
\end{aligned}$$

Fundamentally, the Holevo bound demonstrates that even though n qubits have the capacity to represent a vast amount of classical information due to quantum superposition, the maximum amount of classical information that can actually be extracted from them is limited to n classical bits. Additionally, both theoretical analyses and experimental evidence have shown that during certain computations, quantum bits can transmit more information throughout the computation process than what is achievable using classical systems.

6. QUANTUM CODES

Adapting existing classical methods for quantum error correction is not straightforward, since qubits are subject to the no-cloning theorem, meaning quantum information cannot be duplicated in the same way as classical information. Furthermore, it is not possible to perform arbitrary measurements on a qubit due to the problem of wavefunction collapse. However, we have tools for overcoming these issues thanks to Peter Shor.

6.1. Classical error correction. The basic classical error-correcting code relies on redundancy. For example, the three-bit encoder maps a given alphabet $\mathcal{B}$ into a code alphabet $\mathcal{C}_3$

$$\mathcal{B} = \{0, 1\} \longrightarrow \mathcal{C}_3 = \{000, 111\},$$

where 000 and 111 are called logical codewords for $\mathcal{C}_3$. This code ensures some reliability against bit-flip errors. In particular, if there is at most 1 bit-flip error during transmission (e.g. '0' gets mapped to '001'), the receiver will still be able to recover the original information by a majority vote. If there is a probability p of having a bit-flip error for every bit independently, then

$$\mathbb{P}(\text{error}) = 1 - \mathbb{P}(0 \text{ bit-flip error}) - \mathbb{P}(1 \text{ bit-flip error}) = 1 - (1 - p)^3 - 3p(1 - p)^2 = 3p^2 - 2p^3$$

and

$$3p^2 - 2p^3 < p \iff p \in \left[0, \frac{1}{2}\right).$$

6.2. Quantum error correction. Unlike classical bits, qubits assume a continuum of values. We can write a qubit as

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle$$

By construction, qubits are subject to different types of errors. We can imagine them lying on the Bloch sphere. Thus, they can rotate from one point to another. Mathematically, errors are represented by unitary operations $U(\delta\theta, \delta\phi)$ such that

$$U(\delta\theta, \delta\phi)|\psi\rangle = \cos \frac{\theta + \delta\theta}{2} |0\rangle + e^{i(\phi + \delta\phi)} \sin \frac{\theta + \delta\theta}{2} |1\rangle$$

Luckily, it turns out that quantum errors can be **digitized** so that the ability to correct for a finite set of errors is sufficient to correct for any error. We now explore why.

The single-qubit coherent error process $U(\delta\theta, \delta\phi)$ can be expanded as follows:

$$U(\delta\theta, \delta\phi)|\psi\rangle = \alpha_I \text{Id}|\psi\rangle + \alpha_X X|\psi\rangle + \alpha_Z Z|\psi\rangle + \alpha_Y Y|\psi\rangle$$

where $\alpha_{I,X,Y,Z}$ are the expansion coefficients. By noting that the Pauli Y -matrix is equivalent (up to a phase) to the product XZ , this expression can be further simplified to

$$U(\delta\theta, \delta\phi)|\psi\rangle = \alpha_I \text{Id}|\psi\rangle + \alpha_X X|\psi\rangle + \alpha_Z Z|\psi\rangle + \alpha_{XZ} XZ|\psi\rangle.$$

6.3. Quantum error types. We can divide quantum errors as X and Z -errors. We can think of X -errors as quantum bit-flip errors and Z -errors as quantum phase-flip errors. For the latter case, there is no classical equivalent.

X -type errors: $X|\psi\rangle = \alpha X|0\rangle + \beta X|1\rangle = \alpha|1\rangle + \beta|0\rangle$.

Z -type errors: $Z|\psi\rangle = \alpha Z|0\rangle + \beta Z|1\rangle = \alpha|0\rangle - \beta|1\rangle$.

Digitisation implies that it is possible to use certain techniques from classical coding theory in quantum error correction. However, there are three complications:

- The **no-cloning theorem** for quantum states, meaning it does not exist U_{clone} such that

$$U_{\text{clone}}(|\psi\rangle \otimes |0\rangle) \rightarrow |\psi\rangle \otimes |\psi\rangle$$

- Qubits are susceptible to **two types of errors** at the same time.
- **Wavefunction collapse.**

In quantum codes, redundancy is added by expanding the Hilbert space in which the qubits are encoded. We now describe the two-qubit code, a prototypical quantum code designed to detect a single-bit flip error. The encoding stage of the two-qubit code, acting on the general state $|\psi\rangle$, has the following action:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \xrightarrow{\text{two-qubit encoder}} |\psi\rangle_L = \alpha|00\rangle + \beta|11\rangle = \alpha|0\rangle_L + \beta|1\rangle_L,$$

This does not correspond to cloning the state, as:

$$|\psi\rangle_L = \alpha|00\rangle + \beta|11\rangle \neq |\psi\rangle \otimes |\psi\rangle.$$

The effect of the encoding operation is to distribute the quantum information in the initial state $|\psi\rangle$ across the entangled two-party logical state $|\psi\rangle_L$. This introduces redundancy to the encoding that can be exploited for error detection. Before encoding, the single qubit is parametrised within a two-dimensional Hilbert space

$$|\psi\rangle \in \mathbf{H}_2 = \text{span}\{|0\rangle, |1\rangle\}.$$

After encoding, the logical qubit lies in a four-dimensional Hilbert space

$$|\psi\rangle \in \mathbf{H}_4 = \text{span}\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}.$$

More specifically, the logical qubit is defined within a two-dimensional subspace of this expanded Hilbert space

$$|\psi\rangle_L \in \mathcal{C} = \text{span}\{|00\rangle, |11\rangle\} \subset \mathbf{H}_4,$$

where $\mathcal{C}$ is called the codespace. Now, imagine that the logical qubit is subject to a bit-flip error on the first qubit resulting in the state

$$X_1|\psi\rangle_L = \alpha|10\rangle + \beta|01\rangle \in \mathcal{F} = \text{span}\{|10\rangle, |01\rangle\} \subset \mathbf{H}_4,$$

where we call $\mathcal{F}$ the error subspace. Notice that an X_2 -error will also rotate the logical state into the $\mathcal{F}$ subspace. If the logical state $|\psi\rangle_L$ is uncorrupted, it occupies the codespace $\mathcal{C}$, whereas if it has been subject to a single-qubit bit-flip, it occupies the error space $\mathcal{F}$. As the $\mathcal{C}$ and $\mathcal{F}$ subspaces are mutually orthogonal, it is possible to distinguish which subspace the logical qubit occupies via a projective measurement without

compromising the encoded quantum information. In the context of quantum coding, measurements of this type are called **stabilizer measurements**.

The projective measurement performed is $Z_1 Z_2$, which is said to stabilize $|\psi\rangle_L$ since it leaves it unchanged, but projects errored states $X_1|\psi\rangle_L$ and $X_2|\psi\rangle_L$ onto the (-1) eigenspace.

The two-qubit code works under a more general probabilistic error. Assume that each qubit is subject to a coherent error of the form

$$\epsilon = \alpha_I \text{Id} + \alpha_X X$$

We see that $|\alpha_X|^2 = p_X$ is the probability of an X -error occurring on the qubit. The probability of no error occurring is therefore equal to $|\alpha_I|^2 = 1 - p_X$.

Then the action of the error operator $\mathcal{E}$ acting on both qubits is given by

$$E = \mathcal{E}_1 \otimes \mathcal{E}_2 = \alpha_I^2 \text{Id}_1 \text{Id}_2 + \alpha_I \alpha_X (X_1 + X_2) + \alpha_X^2 X_1 X_2.$$

With the above error operator E , the syndrome extraction stage transforms the quantum state as follows:

$$E|\psi\rangle_L|0\rangle_A \xrightarrow{\text{syndrome extraction}} (\alpha_I^2 \text{Id}_1 \text{Id}_2 + \alpha_X^2 X_1 X_2)|\psi\rangle_L|0\rangle_A + \alpha_I \alpha_X (X_1 + X_2)|\psi\rangle_L|1\rangle_A.$$

6.4. The three-qubit error correction code. The syndrome produced by the two-qubit code informs of the presence of an error but does not provide enough information to allow us to infer which qubit the error occurred on. It is therefore a **detection code**. In order to create an **error correction code** with the ability to both detect and localise errors, multiple stabilizer measurements need to be performed.

We now describe the **three-qubit code**, the natural extension of the two-qubit code in which the encoding operation distributes the quantum information across an entangled three-party state to give a logical state of the form $|\psi\rangle_L = \alpha|000\rangle + \beta|111\rangle$. It lies in an eight-dimensional Hilbert space that can be partitioned into four two-dimensional subspaces as follows:

$$\mathcal{C} = \text{span}\{|000\rangle, |111\rangle\}, \quad \mathcal{F}_1 = \text{span}\{|100\rangle, |110\rangle\},$$

$$\mathcal{F}_2 = \text{span}\{|010\rangle, |101\rangle\}, \quad \mathcal{F}_3 = \text{span}\{|001\rangle, |110\rangle\}.$$

each single-qubit error produces a unique two-bit syndrome $S = s_1 s_2$, enabling us to choose a suitable recovery operation.

To differentiate between these subspaces, we perform two stabilizer measurements $Z_1 Z_2$ and $Z_2 Z_3$ via the circuit shown below, along with the resulting syndromes.

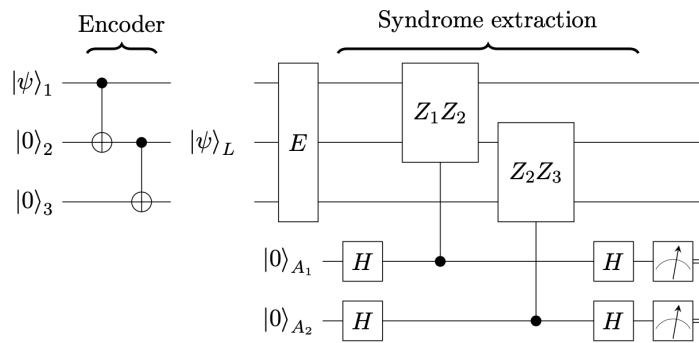


FIGURE 1. The circuit diagram. Encode stage: The information contained in a single qubit is entangled with two redundancy qubits. The stabilizers $Z_1 Z_2$ and $Z_2 Z_3$ are measured on the logical qubit via two operations controlled on the ancilla qubits A_1 and A_2 .

Error	Syndrome, S	Error	Syndrome, S
$I_1 I_2 I_3$	00	$X_1 X_2 I_3$	01
$X_1 I_2 I_3$	10	$I_1 X_2 X_3$	10
$I_1 X_2 I_3$	11	$X_1 I_2 X_3$	11
$I_1 I_2 X_3$	01	$X_1 X_2 X_3$	00

TABLE 1. The syndrome table for all bit-flip errors. The syndrome S is a two-bit string formed by concatenating the results of the two stabilizer measurements.

6.5. Quantum code distance. The distance d of a quantum code is the minimum weight of a Pauli error that can cause a logical error in the encoded information without being detectable by the syndrome measurement. In other words, it is a measure of how many physical qubits an error must affect before it becomes undetectable (or uncorrectable) by the code. It is important for:

- **Error Detection:** A code of distance d can detect any error that affects fewer than d qubits.
- **Error Correction:** A code can correct errors up to $\lfloor (d-1)/2 \rfloor$ qubits, meaning that for high-distance codes, more errors can be corrected.
- **Fault Tolerance:** Larger code distances provide better protection against decoherence and noise, which is critical for implementing reliable quantum computation.

6.6. Stabilizer codes. The three-qubit code is designed such that if an X-error occurs, the logical state is rotated to an orthogonal error space. This event can be detected via a sequence of two stabilizer measurements. We now describe how to generalise this to create $[[n,k,d]]$ stabilizer codes, where n is the total number of qubits, k is the number of logical qubits and d is the code distance.

Here there are k data qubits entangled with $m = n - k$ redundancy qubits. Thus, data is distributed across the expanded Hilbert space. Errors can then be detected by performing m stabilizer measurements P_i . Each of the stabilizers is measured using the same syndrome extraction method that was used for the two-qubit code. If the stabilizer commutes with an error E the measurement of ancilla qubit A_i returns '0'. Thus, the task of constructing a good code involves finding stabilizers that anti-commute with the errors to be detected.

6.7. Example: The Shor $[[9,1,3]]$ code. The first quantum error-correcting code was proposed by Peter Shor in 1995. The Shor code, denoted as the $[[9, 1, 3]]$ code, encodes one logical qubit into nine physical qubits and can correct arbitrary single-qubit errors. If only one of the nine physical qubits suffers any error (bit-flip, phase-flip, or both) the Shor code can fully recover the logical qubit.

6.7.1. Encoding. The Shor code protects against both bit-flip (X) and phase-flip (Z) errors by utilizing both repetition and entanglement. The logical qubit states are encoded as follows:

$$|0\rangle_L = \frac{1}{2\sqrt{2}} (|000\rangle + |111\rangle) \otimes (|000\rangle + |111\rangle) \otimes (|000\rangle + |111\rangle),$$

$$|1\rangle_L = \frac{1}{2\sqrt{2}} (|000\rangle - |111\rangle) \otimes (|000\rangle - |111\rangle) \otimes (|000\rangle - |111\rangle).$$

This encoding consists of:

- A *bit-flip code* using three physical qubits to protect against bit-flip errors.
- A *phase-flip code* using three groups of three encoded qubits to protect against phase-flip errors.

6.7.2. *Error Detection and Correction.* The Shor code can detect and correct arbitrary single-qubit errors by measuring stabilizers. The bit-flip errors are detected using **majority voting** within each triplet. Similarly, phase-flip errors are detected using parity checks across the triplets.

The stabilizers of the above code are given by

$$\mathcal{S}_{[[9,3,3]]} = \langle Z_1 Z_2, Z_2 Z_3, Z_4 Z_5, Z_5 Z_6, Z_7 Z_8, Z_8 Z_9, X_1 X_2 X_3 X_4 X_5 X_6, X_4 X_5 X_6 X_7 X_8 X_9 \rangle.$$

TABLE 2. The syndrome table for single-qubit X - and Z -errors on the nine-qubit code. The nine-qubit code is a degenerate code, as certain Z -errors share the same syndrome.

Error	Syndrome, S	Error	Syndrome, S
X_1	10000000	Z_1	00000010
X_2	11000000	Z_2	00000010
X_3	01000000	Z_3	00000010
X_4	00100000	Z_4	00000011
X_5	00110000	Z_5	00000011
X_6	00010000	Z_6	00000011
X_7	00001000	Z_7	00000001
X_8	00001100	Z_8	00000001
X_9	00000100	Z_9	00000001

If an error is detected, a correction is applied by flipping the erroneous qubit based on majority voting. The Shor code has *distance* $d = 3$, meaning it can:

- Detect up to *two errors*.
- Correct up to *one arbitrary error*.

The distance is defined as the minimum number of qubits that need to be altered to map one logical state to another. From the Shor $[[9, 1, 3]]$ code, we can construct more advanced quantum error correction techniques, such as concatenated codes and surface codes (e.g. $[[5, 1, 2]]$).

6.8. Practical considerations.

6.8.1. *Efficient Decoding Algorithms.* Given a code syndrome S , the role of the decoder is to find the best recovery operation $\mathcal{R}$ to restore the encoded quantum information to the codespace. In stabilizer codes, measuring the stabilizers of an $[[n, k, d]]$ code produces an m -bit syndrome, where $m = n - k$. This means there are 2^m possible syndromes, making decoding computationally expensive for large codes. For small codes, it is possible to precompute **lookup tables** that list the best recovery operation for each of the 2^m syndromes. However, such a strategy rapidly becomes impractical as the code size increases. For large-scale quantum error correction codes we usually use **approximate inference techniques** to determine the most likely error that has occurred given a certain syndrome. One commonly used method is *minimum weight perfect matching* (MWPM), which identifies error chains between positive syndrome measurements. However, there is no known universal decoder that can efficiently be applied to all quantum error correction codes. Different codes require different decoding strategies.

6.8.2. *Code Thresholds.* A crucial metric in QECC is the **threshold theorem**, which states that increasing the distance d of a code can reduce the logical error rate p_L , provided that the physical error rate p of individual qubits remains below a threshold $p < p_{\text{th}}$. This suggests that quantum error correction can, in principle, arbitrarily suppress errors if p is small enough.

6.8.3. *Fault Tolerance*. In real-world quantum circuits, errors can occur not only in specific locations but throughout the circuit, including encoding, syndrome extraction, and recovery operations. This means that an effective QECC must be **fault tolerant**, ensuring that errors in one part of the system do not spread uncontrollably and corrupt the encoded quantum information.

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